

Jeongmin Bae - Portfolio

+61-421-530-328 | jaymin.bae@gmail.com

[in jeongminb](#) | [jeongminb](#) | [jeongminb.github.io](#)

OVERVIEW

My research focuses on spatial representation learning from real-world visual data, including dynamic 3D/4D reconstruction, Gaussian Splatting, NeRF-based scene modeling, egocentric hand reconstruction, and robust reconstruction from imperfect observations.

I am currently interested in developing 3D/4D reconstruction and scene understanding methods that work well in less controlled real-world settings, where visual observations often include challenges such as incomplete data, noise, occlusions, unsynchronized videos, and moving cameras.

At the University of Melbourne, I am currently a Postdoctoral Research Fellow leading a mixed-reality project on real-world scene reconstruction using multiple 360° camera–LiDAR sensor pairs. I received my Ph.D. in Artificial Intelligence from Yonsei University in February 2026.

ONGOING RESEARCH DIRECTION

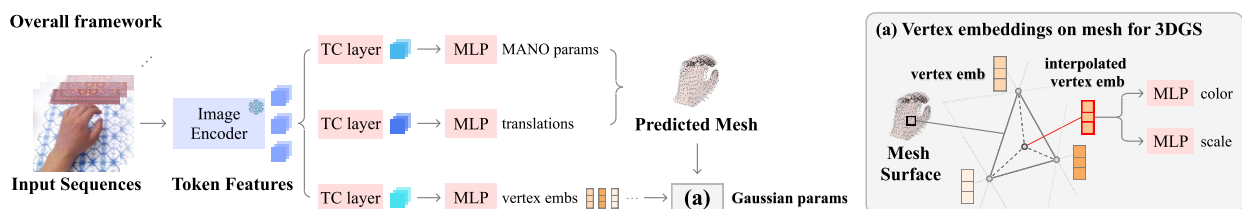
Hand-4DGS: Feed-Forward 3D Gaussian Splatting for 4D Hand Reconstruction from Egocentric Videos

Ongoing work, *arXiv 2026*, Co-first author

Problem Definition

- Reconstructing 4D hands from egocentric videos is important for applications such as AR/VR, AI glasses, robot manipulation, and teleoperation.
- Egocentric hand reconstruction is challenging due to single-view depth ambiguity, rapid hand motion, severe occlusion, and moving cameras.
- Existing methods often rely on multiple viewpoints, which limits their applicability to wearable and egocentric settings. Per-video optimization methods are also too slow for real-time applications.
- This motivates a feed-forward framework that can rapidly reconstruct 4D hands from single-view egocentric videos and generalize to unseen videos.

Approach



- I introduced a mesh-guided 3D hand representation that uses the MANO hand mesh as a structural prior and assigns learnable embeddings to mesh vertices.
- The method samples Gaussians on the mesh surface and interpolates vertex embeddings to define the geometry and appearance of each Gaussian.
- It uses vision foundation model features together with temporal convolution layers to aggregate information from adjacent frames and reduce temporal jitter.
- Outputs from an existing hand pose estimation model are used as initial supervision, while image-level reconstruction losses help refine the hand appearance and pose during training.
- This feed-forward pipeline enables instant 4D hand reconstruction from RGB video inputs without per-video optimization.

Results

- Implemented a feed-forward hand reconstruction pipeline capable of real-time inference and more stable reconstruction than per-scene optimization baselines. (Inference speed: ~60 FPS; EVA PSNR: 23.56 → Ours: 24.66.)
- Demonstrated instant 4D hand reconstruction from RGB inputs on unseen videos.
- Reduced inference-time dependency on pose estimators while improving hand pose and appearance reconstruction through image-level supervision.

E-D3DGS: Per-Gaussian Embedding-Based Deformation for Dynamic 3D Gaussian Splatting

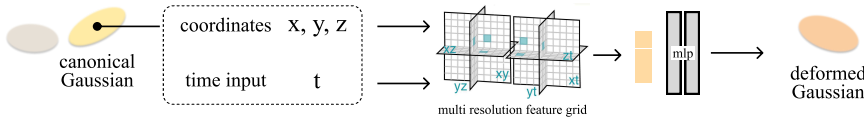
ECCV 2024, Co-first author

Problem Definition

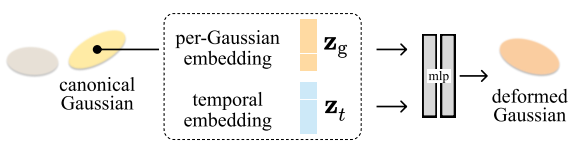
- Existing deformable 3D Gaussian Splatting methods often produce blurry results or fail to accurately represent dynamic regions in complex scenes.
- Many methods predict temporal changes from the coordinates of Gaussian primitives in canonical space, which can make it difficult to model different motions for nearby scene elements.
- As scene dynamics become more complex, these approaches become increasingly dependent on the capacity of MLPs or the resolution of feature grids, increasing model complexity and memory cost.
- This motivates a dynamic scene representation that can effectively model complex motion without relying heavily on deep networks or high-resolution grid representations.

Approach

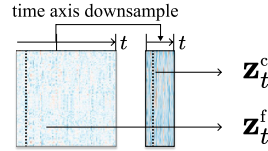
Previous methods



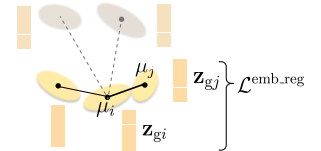
Our method



Temporal embedding



Local smoothness regularization



- I proposed an embedding-based dynamic scene representation that assigns an individual learnable latent embedding to each Gaussian primitive.
- A per-Gaussian latent embedding captures Gaussian-specific deformation, while a global temporal embedding represents the scene state at each time step.
- The deformation of each Gaussian is predicted by a shallow decoder using its per-Gaussian embedding and the temporal embedding for the corresponding time step.
- I used a coarse-fine deformation strategy to separate large-scale motion from fine-grained local details by dividing temporal embeddings into coarse and fine embeddings.
- I also introduced local smoothness regularization based on the assumption that nearby Gaussians tend to undergo similar deformations, helping preserve texture details in dynamic regions.

Results

- On the Neural 3D Video Dataset, the proposed method improved reconstruction quality, rendering speed, and model efficiency compared to 4D Gaussians. (PSNR: 30.71 $\rightarrow$ 31.31; Rendering speed: 51.9 FPS $\rightarrow$ 74.5 FPS; Model size: 59 MB $\rightarrow$ 35 MB.)
- The method also achieved better results than baselines in monocular video settings, demonstrating robust performance across different capture setups.
- This work was published at ECCV 2024, and related patent applications were filed in Korea and the United States (KR 10-2024-0043684 / US 2025/0308078 A1).

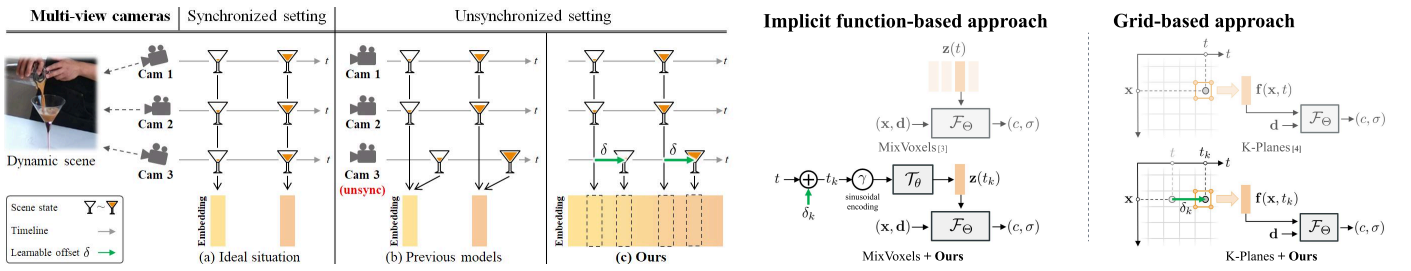
Sync-NeRF: Generalizing Dynamic NeRFs to Unsynchronized Videos

AAAI 2024, Co-first author

Problem Definition

- Existing multi-view dynamic reconstruction methods often assume that images with the same frame index capture the same moment in time.
- In practice, capture timing can differ across cameras. In a commonly used multi-view dynamic video dataset, some views were significantly misaligned despite being captured with synchronization equipment such as timecode systems.
- Training with unsynchronized views can cause ghosting and artifacts in dynamic regions, which degrades reconstruction quality.
- This motivates a dynamic scene reconstruction method that can handle multi-view capture data with timing misalignment.

Approach



- I co-proposed a training strategy that directly compensates for camera-specific time offsets instead of discarding unsynchronized multi-view videos.
- The method introduces a learnable time offset for each camera and optimizes it jointly with the dynamic NeRF representation.
- This allows the model to align observations from different cameras while learning a time-dependent 3D scene representation.
- The strategy is compatible with both implicit-function-based temporal embeddings and grid-based temporal representations.

Results

- The proposed method improved reconstruction quality across multiple baselines under unsynchronized multi-view settings. (MixVoxels: PSNR 29.96 $\rightarrow$ 30.53; K-Planes: PSNR 29.16 $\rightarrow$ 30.44.)
- The learned time offsets compensated for capture-time misalignment, enabling robust dynamic reconstruction from multi-view videos with timing errors.
- This work was published at AAAI 2024, and related patent applications were filed in Korea and the United States (KR 10-2023-0105173 / US 2025/0054224 A1).

OTHER PUBLICATIONS

- 4D Scaffold Gaussian Splatting with Dynamic-Aware Anchor Growing for Efficient and High-Fidelity Dynamic Scene Reconstruction**, AAAI 2026. Introduces a 4D anchor-based framework that reduces storage cost while maintaining high visual quality and fast rendering speed for dynamic scene reconstruction.
- Compensating Spatiotemporally Inconsistent Observations for Online Dynamic 3D Gaussian Splatting**, SIGGRAPH 2025. Addresses temporally inconsistent reconstruction in online dynamic 3DGS by introducing a residual map.
- Rethinking Open-Vocabulary Segmentation of Radiance Fields in 3D Space**, AAAI 2025. Proposes an evaluation protocol for open-vocabulary 3D segmentation that jointly assesses geometry and segmentation quality.
- BallGAN: 3D-aware Image Synthesis with a Spherical Background**, ICCV 2023. Learns foreground 3D geometry from image collections by modeling the background as a 2D sphere.
- FurryGAN: High Quality Foreground-aware Image Synthesis**, ECCV 2022. Trains a GAN to separate fine-detailed foregrounds from single-view image collections without external models or additional supervision.